

Adding the Play Button

In this section, we'll add a Play button that appears before a game begins and reappears when the game ends so the player can play again.

Right now, the game begins as soon as you run *alien_invasion.py*. Let's start the game in an inactive state and then prompt the player to click a Play button to begin. To do this, modify the `__init__()` method of `AlienInvasion`:

```
alien_invasion.py    def __init__(self):
                    """Initialize the game, and create game resources."""
                    pygame.init()
                    --snip--

                    # Start Alien Invasion in an inactive state.
                    self.game_active = False
```

Now the game should start in an inactive state, with no way for the player to start it until we make a Play button.

Creating a Button Class

Because Pygame doesn't have a built-in method for making buttons, we'll write a `Button` class to create a filled rectangle with a label. You can use this code to make any button in a game. Here's the first part of the `Button` class; save it as *button.py*:

```
button.py    import pygame.font

            class Button:
                """A class to build buttons for the game."""

                ❶    def __init__(self, ai_game, msg):
                    """Initialize button attributes."""
                    self.screen = ai_game.screen
                    self.screen_rect = self.screen.get_rect()

                    # Set the dimensions and properties of the button.

                    ❷    self.width, self.height = 200, 50
                    self.button_color = (0, 135, 0)
                    self.text_color = (255, 255, 255)

                    ❸    self.font = pygame.font.SysFont(None, 48)

                    # Build the button's rect object and center it.

                    ❹    self.rect = pygame.Rect(0, 0, self.width, self.height)
                    self.rect.center = self.screen_rect.center

                    # The button message needs to be prepped only once.

                    ❺    self._prep_msg(msg)
```

First, we import the `pygame.font` module, which lets Pygame render text to the screen. The `__init__()` method takes the parameters `self`, the `ai_game` object, and `msg`, which contains the button's text ❶. We set the button dimensions ❷, set `button_color` to color the button's rect object dark green, and set `text_color` to render the text in white.